

Равномерное распределение и тригонометрическая динамика

Домашнее задание

1 сентября 2026 года

Теоретический минимум. Equidistribution modulo 1 and Weyl's criterion; Kronecker's density theorem; the Riemann–Lebesgue lemma and averaging of fast oscillations; the circle rotation and unique ergodicity; the doubling map and the arcsine law; Abel and Dirichlet tests for series with oscillating factors.

1. Prove that $(\{n^2\alpha\})$ is equidistributed for irrational α (Weyl), and more generally that $(\{p(n)\})$ is equidistributed for any polynomial p with at least one irrational coefficient besides the constant term.
2. (Мехмат, 2001) Let $\{\alpha_n\}$ be real numbers such that the functions $\sin(\alpha_n x)$ converge on a set of positive Lebesgue measure. Prove that $\{\alpha_n\}$ has a finite limit.
3. (RMO) Let $f: [-1, 1] \rightarrow \mathbb{R}$ be continuous. Show that
 - a) if $\int_0^1 f(\sin(x + \alpha)) dx = 0$ for every $\alpha \in \mathbb{R}$, then $f \equiv 0$;
 - b) if $\int_0^1 f(\sin(nx)) dx = 0$ for every $n \in \mathbb{Z}$, then $f \equiv 0$.
4. Prove that $\frac{1}{N} \sum_{n \leq N} |\sin n| \rightarrow \frac{2}{\pi}$ and $\frac{1}{N} \sum_{n \leq N} \sin^2 n \rightarrow \frac{1}{2}$, and explain why the same argument fails for $\frac{1}{N} \sum_{n \leq N} \frac{1}{|\sin n|}$.
5. (теорема о трёх расстояниях) Prove that the points $\{\alpha\}, \{2\alpha\}, \dots, \{N\alpha\}$ split the circle into arcs of at most three distinct lengths, and that the largest is the sum of the other two.
6. Prove that $\{\log_{10} n\}$ is dense in $[0, 1)$ but not equidistributed, whereas $\{n \log_{10} 2\}$ is equidistributed; deduce Benford's law for the leading digits of 2^n .

7. Prove that $\frac{1}{N} \sum_{n \leq N} f(\{n\alpha\}) \rightarrow \int_0^1 f$ for every Riemann integrable f and irrational α , and give an example of a bounded Lebesgue measurable f for which this fails.
8. Prove that the map $x \mapsto 2x^2 - 1$ of $[-1, 1]$ preserves the measure $\frac{dx}{\pi\sqrt{1-x^2}}$, is conjugate to the doubling of the angle, and has periodic points of every period; describe them through $\cos \frac{2\pi k}{2^n \pm 1}$.
9. Prove that $\sum_{n \geq 1} \frac{\sin n}{n}$ converges but $\sum_{n \geq 1} \frac{|\sin n|}{n}$ diverges, and compute the first sum.
10. Let α be irrational with bounded partial quotients. Prove that $|\sum_{n \leq N} e^{2\pi i n \alpha}| = O(1)$ and that the discrepancy of $(\{n\alpha\})$ is $O(\frac{\log N}{N})$; compare with $\alpha = \pi$, where nothing of the kind is known.